

2 Supplement to MANUSCRIPT_r541

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4 paper/.

5 **S1. The measurement model of §3.5, in full**

6 The figures in §3.5 come from a noise model that has to be stated before they are used. It adds
7 independent Gaussian jitter to each frame, resamples to the stated frame rate, scales the trace
8 by the cosine of a yaw angle, and displaces the gait event by up to one frame. Four of its
9 properties bear on how the results should be read.

10 **It is optimistic.** The jitter is independent and Gaussian, whereas real markerless error is
11 temporally correlated and biased under occlusion; and the yaw term, applied to a planar trace,
12 is a uniform multiplicative scale that cannot change a rank, a refitted threshold or a
13 classification. It is reported here as inert rather than as a stressor.

14 **It exempts event detection.** The swing window is taken from the noise-free ground reaction
15 force and only perturbed by one frame. A real pipeline estimates it: against a force-plate
16 standard, kinematic algorithms place 94 per cent of treadmill events within one frame in
17 healthy subjects but only 89 per cent within *two* frames in impaired populations [23].

18 **It models one recording, not two.** The error that governs whether a patient's reading can be
19 interpreted (marker replacement, recalibration, day-to-day variation) does not average down
20 over cycles and is the 1.77° figure of §3.5 rather than anything in this grid.

21 **The modality matters more than the model.** Published sagittal errors for markerless video
22 are, for the knee, 5.6° MAE in unimpaired adults [21] and under 4° post-stroke [22]. **For the**
23 **ankle, which is this paper's endpoint, they are worse:** 7.4° MAE with a cross-correlation of
24 0.74 to 0.78 against the knee's 0.97 to 0.98 [21], and 6.3° post-stroke at the ankle from sagittal
25 video [22], comparable to the whole 8.731° effect and larger than the 4.14° tone series.

Video-based pose estimation cannot support the grading proposed here, whatever it can do for the knee, and marker-based capture is the only modality these results transfer to.

An earlier application of the same model to the knee trace, across 216 conditions, gave leave-one-run-out accuracy between 0.759 and 1.000 (median 0.940), with run-level disjointness falling below 95 per cent at $\sigma = 2^\circ$ even where accuracy was 97 per cent. **Those figures rest on ten cells from two chains** and 200 noise realisations, so one cell changing side moves them by 10 per cent; §3.5's table rests on all 65 cells from twelve chains at 60 realisations. The two should not be read as one analysis.

The scoring conventions of the §3.5 table. Every column is computed on the full primary dataset, all 23 hyperreflexia, 36 weakness and 6 control runs from eleven chains plus control, over 60 noise realisations per condition. The two detection columns are leave-one-run-out and the group column is leave-one-condition-out. No run is scored by a threshold it helped set, although in the detection columns the other runs of its own chain do enter that threshold, so those two columns are held out at run level and not at chain level; since runs within a chain are not independent (§2.4), that is the more permissive of the two schemes.

The two conventions of the §3.5 table. The detection columns score each cell against a threshold built from the control runs and from the other cells of its own group, never from itself. An earlier version of this analysis fitted that threshold on the cells it then classified, which returns 1.000 by construction for any separated group and inflated the weakness column by 0.19 to 0.27. The correlations are signed, so a reversal counts against a variable and not for it, which is why the weakness column changes sign between the reported endpoint and the others. Those two detection columns are sensitivities and carry no false-positive rate, because no control cell is classified in producing them; the balanced figure that does classify the six controls is 0.687 for the endpoint at 3° .

The tone score. Taking T as the control mean minus the patient's reading, the five hyperreflexia doses give $T = 7.245 \pm 0.064$, 8.331 ± 0.079 , 10.200 ± 0.101 , 10.578 ± 0.293 and $11.381 \pm 0.146^\circ$ on noise-free traces, and the weakness group gives T below 0.886° . The weakness series of 0.66° (Supplement S14) sits between the between-session error of 1.77° and the within-session error of 0.32° : resolvable in principle between two conditions recorded in one session, and not resolvable against a normative reference collected separately, which is the comparison a clinic makes. All five values stay in order at every noise level.

The two published measurement errors, and what their sources permit. Ref [15] publishes two errors for peak swing dorsiflexion, both from treadmill walking and both inverted here through that paper's own definition $MDC = SEM \times 1.96 \times \sqrt{2}$: a between-session error of 1.77° , from sessions twenty days apart and so including marker replacement and day-to-day variation, and a within-session error of 0.32° , across strides of one recording. Its authors state that neither may be used to interpret overground studies. That same table publishes between-session errors for three further sagittal variables that are members of the 81 scanned in §3.4, namely ankle angle at contact at 2.53° , peak knee flexion in swing at 2.06° and hip angle at toe-off at 4.15° ; §3.4 reports how the counts move when each variable is given its own error rather than the smallest of the four. The value used for knee angle at contact comes from a study that defines its standard error of measurement from the change-score SD, which already contains a factor of $\sqrt{2}$. We therefore did not take that paper's published 1.55° . We recomputed the error from its own reported limits of agreement for knee angle at initial contact, -6.78° to $+7.74^\circ$, which give a change-score SD of 3.70° and a single-measurement error of $3.70/\sqrt{2} = 2.62^\circ$ [13]. The two figures are not related by a transformation; they are two derivations, and we use the second. Raising the knee comparator's error from 1.55° to 2.62° lowers that comparator's standardised effect and raises its misassignment rate, so this choice works in this study's favour and we flag it for that

reason. A systematic review of three-dimensional kinematic reliability concludes that errors of this size, while "probably acceptable", are "generally not small enough to be ignored during clinical data interpretation" [24].

What the five columns of §3.5 do and do not measure. The two detection columns are sensitivities and not accuracies: no control cell is classified in producing them. However, they carry no false-positive rate and cannot be read against a chance level, which is undefined without one. A value of 0.512 says that half of isolated weakness runs clear a threshold, not that the variable performs at chance. The balanced figure, which does classify the six control runs, is 0.687 for the reported endpoint at 3° of jitter. Detecting an isolated weakness against an unimpaired limb returns 0.50 to 0.65 across the four candidates tested, which is §3.4's asymmetry expressed as a detection problem rather than a grading one.

The three weakness-grading variables under the noise model. Their rank correlations with weakness depth fall from 0.822, 0.900 and 0.820 on noise-free traces to 0.707, 0.693 and 0.682 at 3° of per-frame jitter, all below the 0.8 criterion §3.4 applies, over 200 realisations of the same model.

A convention note on the ρ column. It was computed with the helper described in §3.4 and Supplement S8, which did not use mid-ranks on a tied dose vector. Correcting that moves the tone correlations by up to 0.12 in the direction of a stronger relation. The column as printed is a lower bound on magnitude, which is the conservative direction for the argument it supports.

S2. Dose response at the contact instant, and the axis spans

The registered prediction that bears on apportionment was made and tested at the contact instant, and the asymmetry is visible there too. Knee angle at contact grades reflex gain monotonically ($r = -0.993$ at unweakened depth, rising 3.437° across the primary series; warm-start chaining, in which each severity level is seeded from the previous level's optimum accumulates about 0.15° per round and over four rounds could account for roughly 17% of

that rise). The ankle at contact grades gain as well ($r = -0.981$) but crosses the weakness band while doing so, which is why the pooled ankle-at-contact difference is near zero: a crossing, not a convergence. Over five stance sub-phases the two lesions displace the knee in **opposite** directions at initial contact ($+1.358^\circ$ weakness, -3.922° hyperreflexia) and loading response, and in the same direction from midstance on. Scored over the three **registered** gain levels against the 0.511° floor of Supplement S10, knee angle at contact spans 8.3 seed SD along the hyperreflexia axis at unweakened depth and 6.9 at $\times 0.80$, against 0.9 to 3.5 along the weakness axis: reaching 3.5 only at the highest registered gain and sitting at the floor below it.

The prediction held at the knee, and at the ankle in swing it depends on which pairs are counted. It predicted that different lesion combinations would produce overlapping readings, and at the knee six registered pairs do, the closest differing by 0.302° , less than the artefact floor. At the reported endpoint the 96 pairs differing in reflex gain do not overlap, the closest differing by 0.833° or 9.3 times that endpoint's mixed-grid floor of 0.089° . The grid supplies 120 pairs, and of the 24 that hold gain fixed and vary dorsiflexor strength alone, four fall below the floor, the closest at 0.044° . An earlier version of this section reported that no pair fell within one seed SD, which was computed on the 96 without saying so; that claim is withdrawn. What the endpoint fails to separate is two limbs whose calves are equally overactive and whose dorsiflexors differ, which is §3.4's asymmetry stated one pair at a time. The two endpoints command almost the same excursion along the gain axis, 8.210° against 8.113° over the grid's range of 0.025 to 0.220, wider than the primary window, which is why both exceed the 3.437° above, on noise floors differing about sixfold. The apportionment fails at the knee for want of resolution, not for want of signal, and neither endpoint rescues the weakness axis (§3.4).

S3. The rectification measurement behind §3.5

§3.5 reports that the group difference compresses under measurement noise. This section gives that measurement. The five tone scores move under noise, and they do not move together. A maximum rectifies noise upward, and the two groups' swing peaks are not equally shaped: the hyperreflexia peak's curvature is 0.158 against the control's 0.251 per sample squared, and 10.3 of its samples lie within half a degree of the maximum against the control's 4.0. The two were measured separately and the second is not derived from the first: a quadratic peak of that curvature would admit about 5 samples, not 10.3. Therefore, the hyperreflexia peak is flatter than a quadratic near its maximum. Either way it recruits more samples into the maximum and therefore rectifies harder. Measured over 250 realisations for the biases and 50 for the tone scores, per-frame jitter of 1, 2 and 3° biases the reading upward by 1.008, 2.397 and 3.897° in the control group, by 1.030, 2.432 and 3.960° in the weakness group, and by 1.477, 3.323 and 5.260° in the hyperreflexia group. The group difference is set by the second and third of those, not by the control, and compresses from 8.731 to 8.284, 7.840 and 7.431°: to 85 per cent of its clean value at 3°. The five T values fall correspondingly, by up to 1.505°, which exceeds three of the four gaps between adjacent levels (1.086, 1.869, 0.378 and 0.803°).

S4. The four remaining controls of §3.3, and the faller selection check

Stationarity. The per-cycle group difference over the five analysed cycles is +8.930, +8.656, +8.790, +8.777 and +8.654°: one sign throughout, with a spread of 0.276° that is 3 per cent of the window mean. Those five average to 8.761° and not the 8.731° reported for the window, because cells contributing a sixth cycle enter the window mean and not the per-cycle series. For comparison the same statistic is 14 per cent for knee angle at contact and 55 per cent for the two-sided ankle band of §3.1, whose first analysed cycle sits 1.4° above the remaining four.

The faller selection check. Because that control turns on end time, the rule choosing one run directory per chain-seed must not itself select on it. Choosing the directory before applying criterion G, as every other analysis in this paper does, and choosing it among the criterion's survivors both return the same 17 runs and the same mean of 9.572° .

The five duration slopes. Fitted within condition, they are -0.006 , -0.005 , $+0.128$, -0.433 and $+0.112^\circ \cdot s^{-1}$. Their mean is -0.041 and the standard error of that mean is 0.102 , which carries $\pm 0.743^\circ$ over the 7.28 s difference in end time between the groups at one standard error, and $\pm 2.06^\circ$ at 95 per cent on five conditions. Every one of the five comes from a hyperreflexia condition, because the rule requires within-condition variation in end time and no weakness condition has any.

Unlesioned side. Both lesions are left-only. On the right ankle the group difference is $+0.047^\circ$, 0.5 per cent of the lesioned side's 8.731° , and the displacement from control by gain is $+0.00$, -0.06 , -0.10 , -0.08 and $+0.01^\circ$ across the five doses. Three of those five exceed the lesion side's 0.036° artefact floor, by up to 2.8 times. However, the unlesioned ankle is not motionless. It is not dose-monotone, and its largest excursion is 1.1 per cent of the lesion side's effect. Knee angle at contact carries 35.5 per cent of its group separation on the unlesioned side. This endpoint carries essentially none, so it registers a lesion-side displacement and not a bilateral re-resolution.

Walking speed is the other candidate confound that turns out not to matter. It differs between the groups by $0.056 \text{ m} \cdot \text{s}^{-1}$, and the within-dose regression of the endpoint on speed has a slope of $-0.563^\circ / (\text{m} \cdot \text{s}^{-1})$, propagating 0.032° , or 0.4 per cent of the effect. Speed is a consequence of the lesion, not an independent variable, so that figure indicates scale and bounds nothing.

Admissibility gate. The gate is 9.73 s. Raising it to 10.50, 11.50, 12.50 and 13.00 s removes 4, 10, 12 and 12 of the 23 hyperreflexia runs and none of the 36 weakness runs, and the run-

172 level gap does not fall: 6.265, 6.275, 6.295, 6.302 and 6.302°. Non-overlap is not an artefact
173 of admitting the shortest-surviving cells.

174 **S5. The chaining measurement behind the artefact floor**

175 Re-running a chain through further optimisation rounds at unchanged severity gives the
176 following cumulative displacements from the chain's start. On the hyperreflexia chain the
177 endpoint moves +0.033, +0.032 and +0.036° after one, two and three further rounds; on the
178 weakness chain it moves +0.018, +0.000 and −0.009°. Neither grows with chain depth, which
179 is why the floor of 0.0358° is used directly and not multiplied by the depth of the deepest
180 level. Knee angle at contact on the same chains drifts −0.035, −0.218 and −0.451° on the
181 hyperreflexia side, roughly 0.15° per round, and −0.586, −0.502 and −0.440° on the weakness
182 side; its 0.586° floor is the largest of those six.

183 **S6. Sub-phase boundaries and the construction of the protocol artefact**

184 The five stance sub-phases of §3.7 divide the interval between successive contacts of the same
185 limb into initial contact, loading response, midstance, terminal stance and pre-swing at the
186 boundaries 0 to 5, 5 to 20, 20 to 35, 35 to 50 and 50 to 70 per cent of the cycle used by [9],
187 which are not the only convention in use. The protocol artefact of §3.3 is constructed by re-
188 running a completed chain through further optimisation rounds at unchanged lesion severity
189 and recording how far the endpoint moves. The largest such displacement over six measured
190 re-runs is 0.586° for knee angle at contact and 0.036° for peak dorsiflexion in swing; S5 gives
191 the six shifts individually.

192 **S7. The preregistration, in full**

193 This section carries the material compressed out of §2.5. It is a record of what was planned. It
194 certifies nothing, and §2.5 says so.

Predictions were registered with hashes before the corresponding cells were run; 55 of the 72 registrations carry verified hashes. Three disclosures qualify that. One registration was edited after its result was produced, together with its analysis script and regenerated outputs, so the provenance block for that analysis cannot match and hash verification cannot detect the breach from the sidecar alone, and the revision the run executed under is not recoverable. The 55 verified registration hashes should be read in that light.

The mixed-grid registration's own designated key prediction is uninformative as registered. That registration fixed a three-column series and named P3 (that a single knee reading recovers the lesion mixture) as the prediction whose failure fails the clinical claim; as registered P3 cannot be evaluated, because its $\times 0.60$ root exists at three seeds and so reaches this paper's four-seed criterion in no cell (§2.3). It is excluded by the seed criterion, not by the criterion, which it passes 10 times in 12. P2, the weakness axis, is likewise uninformative as registered. P4 was informative and held (§3.7): it predicted that different lesion combinations would produce overlapping knee readings, and they do. A fourth gain column and the $\times 0.70$ weakness row were added after hashing and are post hoc; §2.3 gives the grid as it stands and Supplement S10 marks which comparisons rest on them. (An earlier draft of this section described the addition as a fifth weakness row. The archive holds four rows, and the figure given in §2.3 is the archive's.)

Registrations that failed, and the one that produced no result, are not separately listed; the hash audit below covers all 72.

The hash audit. Two audits exist over two different populations and they must not be conflated. Over the 72 registrations: 55 carry a .sha256 sidecar and every one of the 55 still hashes to it. However, there are no mismatched registrations; the other 17 carry no sidecar at all, being pre-r160 or superseded, which is a coverage gap rather than a tampering signal.

Over the 97 sidecars in the deposit directory: 84 verify, 8 do not, 2 further mismatch by

220 design because they are pre-read snapshots taken before a cold read and are expected to
221 diverge from the final file, and 3 cannot be resolved to a target. No sidecar has been rewritten
222 at any point, deliberately: re-hashing would certify the present state and destroy the evidence
223 that an attestation had broken.

224 The chain of custody is broken in one place and it is not a hash. PREREG_2ndbody_r229.md
225 was edited after its results were produced, to restore four outcome levels. The provenance
226 block for that analysis cannot match, because it hashes the registration and the registration
227 changed after the run, and the version the run actually executed is unrecoverable. What
228 survives is the reproduction check: every value in the corresponding deposit is re-derivable
229 from the frozen dataset by the executed code. The breach touches the claim that those values
230 were produced under a registration fixed in advance, and not the values themselves. We
231 record it here because no automated check in this repository would surface it, and a reader is
232 entitled to know that the registration's integrity rests on this disclosure rather than on the
233 hashes.

234 **The four predictions, as registered.** All four were made about knee angle at initial contact
235 on the mixed grid, which is why none of them is a registered test of the endpoint this paper
236 reports. They are translated here from the registration, which is written in Chinese.

237 *P1, the spasticity axis is monotone.* At each level of weakness, the cell mean of knee angle at
238 contact becomes more flexed, that is more negative, as the reflex gain rises, monotonically
239 across three levels.

240 *P2, the weakness axis is monotone and runs the other way.* At each reflex gain, knee angle at
241 contact becomes more extended, that is more positive, as weakness deepens.

242 *P3, mixed lesions can be ordered.* This was designated the key prediction. A fixed threshold
243 exists that separates the spasticity-dominant cells (gain 0.050 with $\times 1.00$ or $\times 0.80$) from the
244 weakness-dominant cell (gain 0.0125 with $\times 0.60$) without overlap. Should they overlap, knee

angle cannot order mixed lesions and the clinical claim fails, which the registration requires be reported as such.

P4, cancellation exists. At least one pair of cells at different combinations of gain and weakness has overlapping ranges of knee angle at contact. A single knee reading cannot uniquely determine the lesion combination. The registration marks this as an expected limitation rather than a failure, and states that if it does not hold, the knee is a fully invertible variable and that is the stronger result.

How each stands. P1 and P2 are uninformative as registered: both are monotonicity statements over three levels on an axis this paper reports over more. P3 cannot be evaluated because its weakness-dominant cell sits on the $\times 0.60$ row, which exists at three seeds and so reaches the registered four-of-six bar in no cell; it is excluded by the seed criterion and not by criterion G, which it passes in 10 of 12. P4 was informative and is tested in §3.7, where it holds at the knee for the six registered pairs and, at the reported endpoint, holds for pairs differing in weakness and fails for pairs differing in tone.

Two further registered controls. The duration control was enforced on P1 to P3, with a propagation above 50 per cent of the effect declaring the result confounded. And a batch null was fixed in advance: the largest boundary difference measured between conditions of the same mechanism was 0.5697° , and no gap between cells smaller than that may be called a separation.

Post hoc elements. Two parts of the mixed grid entered after the registration was hashed: the $\times 0.70$ tibialis anterior row and the highest gain column. Every analysis of the grid uses all 74 admitted runs unless stated otherwise, and The $\times 0.70$ row and the highest gain column are named here so that any comparison resting on them can be identified: §3.7's closest different-gain pair, at 0.833° , is one such comparison, and three of the four sub-floor same-gain pairs are others.

S8. How the counts of §3.4 depend on the threshold and the rank

convention

Two analytic choices move the counts of variables that usably grade each lesion, and §3.4 reports the range they produce rather than a single pair.

The bar. A candidate must span more than one clinical measurement error to be called usable.

Ref [15] Table 2 publishes between-session errors for five sagittal variables, four of them members of the 81: peak ankle angle in swing at 1.77° , ankle angle at contact at 2.53° , peak knee flexion in swing at 2.06° and hip angle at toe-off at 4.15° , each inverted through that paper's own definition $MDC = SEM \times 1.96 \times \sqrt{2}$. An earlier version of §3.4 stated that only two of the 81 variables had published errors, which is false against this source, and applied the smallest of the four to all 81, which is the most permissive choice available. Three thresholds are defensible: the single permissive one, each variable's own error where one exists, and each joint's largest published error.

The rank convention. The helper used to score the ordering computed ranks as the argsort of the argsort, which assigns distinct ranks inside a tie group broken by array order rather than mid-ranks. The dose vector is tied by construction, five gains over 23 hyperreflexia runs and six scalings over 36 weakness runs. However, this is not Spearman's ρ . Correcting it moves the tone correlations by up to 0.12 and the weakness correlations by up to 0.03, in both cases towards a stronger relation. A third reading is available and is the one §3.4's own sentence describes: a condition-level correlation over the five gains or the six scalings, in which no ties exist at all.

bar	rank convention	tone	weakness	ratio
permissive, 1.77° everywhere	as shipped	25	3	8.3
permissive, 1.77° everywhere	Spearman, cell level	28	4	7.0
permissive, 1.77° everywhere	condition level	27	7	3.9
each variable's own error	as shipped	24	3	8.0

each variable's own error	Spearman, cell level	27	4	6.8
each variable's own error	condition level	26	7	3.7
each joint's largest error	as shipped	14	1	14.0
each joint's largest error	Spearman, cell level	17	2	8.5
each joint's largest error	condition level	16	4	4.0

291

292 Sweeping a single common bar from 0.3 to 5.0° instead, the tone count falls from 31 to 2 and

293 the weakness count from 7 to 0. Across the nine combinations of the table the ratio lies

294 between 3.7 and 14.0 and never inverts. The sweep is not covered by that statement: above a

295 threshold of about 3° the weakness count reaches zero while the tone count is still 14.

296 Therefore, the ratio there is unbounded rather than bounded by 14. The counts also index how

297 much of the cycle each lesion is legible over rather than how many independent

298 measurements report it, because the candidates are three curves sampled every 5 per cent plus

299 eighteen landmarks and adjacent samples of one curve are strongly correlated.

300 **S9. The selection-valid permutation and the nested classification**

301 §3.2 reports a permutation p and a leave-one-condition-out accuracy for a variable selected as

302 the maximum of an 81-candidate scan. Both were recomputed with the selection inside the

303 procedure.

304 **Max-*T* permutation.** For each of the $C(11,5) = 462$ assignments of condition labels, the

305 condition-level standardised separation was recomputed for all 81 candidates and the largest

306 taken. The observed maximum is 111.3 seed SD, at peak ankle dorsiflexion in swing. That

307 figure and §3.1's 76.5 share a denominator, the 0.0819° within-condition seed SD, and differ

308 in the numerator: 76.5 divides the run-level gap of 6.265° and 111.3 divides the condition-

309 level mean difference of 9.117°. The permutation uses the condition-level statistic because the

310 unit of analysis is the condition. The null distribution of the maximum has a median of 25.6

and a 95th centile of 67.4, and one arrangement of the 462 attains the observed value, which is the true labelling. The selection-valid p is 0.00216, identical to the naive single-variable figure, because 1/462 is the floor the design can return.

Nested leave-one-condition-out. In each of the eleven folds the winning candidate was chosen using only the ten training conditions, and the classifying threshold fitted on those ten. Peak ankle dorsiflexion in swing was selected in all eleven folds, and the accuracy is 59 of 59.

What the permutation does not show. It cannot return a value below 1/462 on an eleven-condition design and returns exactly that for each of the thirty candidates whose groups are disjoint. It does not discriminate among them. The classification is different and an earlier version of this section said otherwise: because the threshold is fitted on ten conditions, a held-out condition can fall on the wrong side of it even where the groups are globally disjoint. Knee angle at contact is run-disjoint and returns 0.983, not 1.000. The nested classification was run for the winning candidate and not for all thirty. Thus, the claim that it returns 1.000 for every disjoint candidate is withdrawn as unmeasured and, in the one case that was measured, false.

S10. The two corpora, and the four decision rules in full

Lesion conditions outside the primary census.

Five further lesion conditions exist in the archive at 0 of 6 to 5 of 6 admitted. They carry plantarflexor-weakness and bifunctional lesions and are not further hyperreflexia levels, which is also why no plantarflexor-weakness batch null exists. We exclude them from every group comparison and from the scan, and use them once, as the duration-matched control of §3.3, precisely because their lesion is neither of the two under test.

The mixed grid. It is a separate dataset and is not part of the 65-cell primary census summarised in §2.3. It crosses four tibialis anterior scalings ($\times 0.60$, $\times 0.70$, $\times 0.80$, $\times 1.00$) with four delivered reflex gains (0.025, 0.050, 0.100, 0.220) in the same model. The $\times 0.60$ row was

run at three seeds per cell and the other three rows at six, giving $12 + 24 + 24 + 24 = 84$ cells, of which 74 pass the same admissibility gate. The unequal seed count is a property of the archive and was not a design choice, and it means the $\times 0.60$ row contributes half the cells of the others to every grid analysis; where a grid statistic is computed per column or per row it is computed on the cells that exist, and the row is flagged wherever it carries a conclusion. Survival on it is governed by gain and not by weakness: the three lowest gain columns admit 21 of 21 cells each and the highest admits 11 of 21, while admission by weakness row runs 10/12, 21/24, 23/24 and 20/24 from $\times 0.60$ to $\times 1.00$. Two elements of the grid are post hoc and are marked wherever they are used: the $\times 0.70$ row, adopted after the registered $\times 0.60$ root reached the four-seed criterion in no cell, and the highest gain column, added after the registration was hashed. Every analysis of the grid in §3.4 and §3.7 uses all 74 admitted runs unless stated otherwise.

The four decision rules of §2.4. These were fixed before the results were read. Each is named in the text where it is applied, and §3.3 reports the verdict of each.

1. Noise floor. Differences are divided by a within-condition seed SD. Three floors appear in this paper and each is used consistently within its own analysis; where a multiple of "seed SD" is quoted, the divisor is named. The scan of §3.1 and every figure derived from it use a floor computed over the eleven lesion conditions with at least two admitted runs, the control chain excluded, which for knee angle at contact is 0.484° and for peak swing dorsiflexion 0.082° . A uniform rule is necessary because 81 candidates are compared with one another. Excluding the three-cell 0.240 condition, as a four-seed criterion would, lowers the knee floor to 0.425° and would inflate every knee multiple by about 14 per cent, so the floor used here is the conservative one. The mixed-grid analyses use one of two floors and each is named where it is used: 0.511° for the registered-level comparisons of §3.7, built from the subset of grid cells differing solely in seed, and 0.557° for statistics computed over all 74 admitted grid cells.

The peak-swing-dorsiflexion equivalent on the same grid is 0.089° . A reader recomputing a " $\times$ seed SD" figure should use the divisor named beside it, not a global one.

2. Duration control. Simulation end time is propagated across the groups' end-time span; above 50% of the effect the effect is declared confounded.

3. Within-window stationarity. Any endpoint reported as a window mean also reports per-cycle drift against the control group's drift on the same channel.

4. Centring. Displacements are reported against the unlesioned control chain, not between groups alone.

S11. An analysis examined and not reported

An earlier draft reported a third signature of the redistribution described in §3.8: a left-minus-right hip range-of-motion asymmetry, offered as evidence that the effect reaches a joint neither lesioned muscle crosses. It is not reported, for three reasons found on audit. The comparison rested on a single hyperreflexia chain, so no between-condition variance was estimable. Its weakness figure was drawn from an unrecorded three-chain subset where the deposit holds six, whose mean is $+0.479^\circ$ rather than the $+0.45^\circ$ the draft printed. And the edge gap is 1.7 to 3.3 times the seed SD depending on which group's noise threshold is used, rather than the 2.7 to 3.0 stated. The caution in §3.8 rests on the unlesioned-side displacement and does not depend on this analysis.

§3.7's earlier statement that no pair of grid cells fell within one seed SD at the reported endpoint is also not carried in the text as written: it was computed over the 96 pairs differing in reflex gain, and four of the 24 same-gain pairs fall below the grid threshold, the closest at 0.044° . The current text reports all 120.

S12. The limitations of §4.4, with their workings

The model and its lesions

385 No convention governs how spasticity and weakness enter a musculoskeletal simulation.

386 Jansen et al. represent spasticity as increased length and velocity feedback with gains chosen

387 arbitrarily, and state that no claim about absolute effect follows from them [9]; Ong et al.

388 scale maximum isometric force for weakness and shorten tendon slack length for contracture,

389 re-optimising per condition [10]; a further model contrasts plantarflexor hyperreflexia against

390 plantarflexor weakness in two dimensions [11]; and in personalised cerebral-palsy models,

391 impairments of known magnitude explain on average 17 per cent of the kinematic deviation

392 and never more than 39 [12]. How much these representations recover therefore varies widely,

393 and none of them settles the choice.

394 Our study sits inside that spread. Everything here is a property of a single 19-degree-of-

395 freedom model under a single reflex controller, and the asymmetry between the lesions rests

396 on how much reserve this model's tibialis anterior retains in an unloaded swing limb. That is a

397 modelling choice no internal control reaches. We offer the finding as a hypothesis about

398 human gait rather than as a measurement of it.

399 Chisholm et al. report, in 55 independently ambulating sub-acute stroke survivors, that peak

400 dorsiflexion during swing was related to isometric dorsiflexor force at $r = -0.32$, $p = 0.039$

401 [28]: an association in patients, at this study's own endpoint, of the size this model calls faint.

402 Only the published abstract of [28] was retrieved, so the coefficient, the p value and the

403 cohort size are all we can verify. What we cannot check is the one figure that would settle

404 how far the result bites, the cohort's dorsiflexor force distribution. Should it overlap the $\times 0.80$

405 to $\times 0.95$ window across which this model's weakness series spans 0.66° , the model

406 understates the effect of dorsiflexor force on swing dorsiflexion, most plausibly because its

407 tibialis anterior keeps more reserve in an unloaded swing limb than a paretic one does. Should

408 the cohort be weaker than that window, the two results agree and say only that our series stops

short of where weakness becomes legible. The asymmetry should therefore be read as bounded by an unresolved patient result.

A second patient result is often read as contradicting the asymmetry and does not. In 68 chronic hemiparetic adults, dorsiflexor strength was the strongest determinant of gait velocity and of temporal symmetry ($R^2 = 0.30$ and 0.36) while dynamic plantarflexor spasticity was the strongest determinant of spatial symmetry ($R^2 = 0.53$) [27]. Those are spatiotemporal parameters from an instrumented walkway, with no joint kinematics recorded. Weakness may therefore affect timing where it affects sagittal angle little, which is a reasonable reading of the two studies together, although it is a hypothesis this study did not test, since only sagittal joint angles were scanned.

A third pathway by which weakness might still reach the endpoint stays open. If dorsiflexor strength governs walking speed [27] and the endpoint depends on speed at $-0.563^\circ/(\text{m}\cdot\text{s}^{-1})$ (Supplement S4), weakness reaches the endpoint indirectly. The 0.032° figure in §3.3 covers this dataset's $0.056 \text{ m}\cdot\text{s}^{-1}$ group difference; over the 0.2 to $1.2 \text{ m}\cdot\text{s}^{-1}$ range a stroke cohort presents, the same slope propagates about 0.23° across a $0.4 \text{ m}\cdot\text{s}^{-1}$ gap, comparable to the entire direct weakness effect. Simulated *plantarflexor* weakness, a different muscle group again, displaces ankle angle at contact by 15 to 16° [10].

How shallow that range is can be read against the study this method follows. Ong et al. model mild, moderate and severe plantarflexor weakness by reducing maximum isometric force to 25, 12.5 and 6.25 per cent of its original value [10]. The deepest dorsiflexor weakness anywhere in this study, the $\times 0.60$ row of the mixed grid, is 60 per cent, which is more than twice as shallow as their mildest level, and the primary weakness series is shallower still at $\times 0.80$ to $\times 0.95$. We could not go deeper and keep a walking model under this controller, but the consequence is that this study reports what a sagittal angle carries over a narrow band near

the top of the weakness range, and says nothing about the range in which weakness is usually studied or clinically encountered.

We therefore claim the insensitivity to weakness in four respects only, and within them it holds: for tibialis anterior scaling between $\times 0.60$ and $\times 1.00$, for sagittal joint angles, for a design in which walking speed is nearly matched across the groups, and for this model and controller. Outside them it is untested, and inside them it is contested by [28], whose scope §4.4 states.

What the design cannot separate

Losing gait under a sufficiently deep deficit is a property of this class of model rather than of this study: Ong et al. report that theirs walks without substantial adaptation until both plantarflexors are moderately or severely weakened [10], and severity and survival are therefore coupled wherever a predictive simulation is driven to the edge of its stable range. In this dataset the coupling falls along the lesion axis. None of the 23 admitted hyperreflexia runs reaches the simulation horizon and all 36 weakness runs do. Therefore, no group comparison at matched survival exists here, for this variable or any other: the confound belongs to the dataset rather than to the endpoint. Cells of the mixed grid at the two lowest gains do reach the horizon, which is what makes a survival-matched comparison possible there at all, but those gains lie below the primary window and the comparison they support is between two gains rather than between the groups. The within-dose duration regression gives 3 ± 9 per cent of the group effect at one standard error, or 3 ± 24 per cent at 95 per cent on five conditions, and is an extrapolation rather than a bound because no weakness condition contributes to its slope. The seventeen non-reflex fallers of §3.3, dying in the same band with a normal swing ankle, show that short survival is not sufficient for the effect. Both leave open whether the hyperreflexia group's own loss of balance contributes to it, and that group is

laterally divergent inside the measurement window. It is losing balance while it is measured, which the admissibility criterion does not test for.

The two lesions are imposed independently here, and crossing them to obtain independent magnitudes is what makes the apportionment question answerable at all. Patients do not present that way ([4]; Supplement S13). In a correlated population a variable shown here to be insensitive to weakness will still track weakness through its association with tone, so a low swing dorsiflexion cannot be attributed to one component from the reading alone. Since the treatment decision does turn on which component is present, chemodenervation for overactivity against orthosis or transfer for weakness (§1), this limits the reading's clinical use and not merely the external validation of the specificity claim.

Predictive simulations reproduce the gross shape of the ankle trajectory and not its detail: an earlier version of this section stated that the unlesioned model of Ong et al. contacts the ground at 15 to 16° of dorsiflexion. That misread the source: those are the *increases* in dorsiflexion at contact that moderate and severe plantarflexor weakness produce relative to that paper's own unimpaired simulation [10], not an absolute angle, and we can draw no conclusion about that model's baseline from them. The observation below therefore stands on its own without support from the comparator. Our unlesioned ankle reproduces the loading-response plantarflexion trough and the toe-off excursion (−19.8° at 59.5 to 63 per cent of cycle, against a normal −15 to −20° at 60 to 62 per cent), and reaches its swing peak at the phase a normal ankle does. Of four landmarks only that last falls where it should. The loading-response trough is early and shallow, the ankle dorsiflexes to +10.7° by 15 per cent of cycle where a normal one is near neutral, the terminal-stance peak is consequently early, and the swing peak at 9.5° sits above the normal range of 0 to 5° entirely. Readouts in mid-to-late stance, including the two-sided band of §3.1, sit on the descending limb of that non-physiological excursion, and their phase does not correspond to the same event in a patient.

Taking the endpoint in swing avoids the phase problem and not the value problem: every displacement we report is measured from a control whose swing peak is itself outside the normal range.

The status of the result

We chose the endpoint after seeing the data, from a scan of 81 candidates. We apply no correction for multiplicity, and treat no analysis here as confirmatory, and the figures for the endpoint are the maximum of that scan and so an upper estimate of what a pre-specified choice would have found. The statement that survives the selection is §3.4's, which holds across all 81.

A reflex-controlled model with graded deficits of known magnitude and the controller re-optimised per deficit is established [10], as is grading a spindle-feedback gain inside a gait model [9], which also reports the direction of the swing effect this paper measures. What neither did is cross two lesions of different kinds against one another and score a variable on what it says about each.

Every figure bearing on a clinic, the discriminability of §3.5 and the attenuation of §4.4, combines this study's effect sizes with published between-patient SDs and reliabilities. We measured nothing here in a patient.

S13. Why this question needs a simulation

Apportioning an equinovarus foot between calf overactivity and dorsiflexor weakness requires knowing how much of each is present. That is where the difficulty appears to lie, and it lies upstream of any measurement.

In a patient neither quantity is available. The two components are graded on incommensurable ordinal scales (a Tardieu grade against a manual muscle test) and neither is a magnitude. They co-occur: in the cohort of [4], median manual muscle testing of the dorsiflexors was 1 of 5 in

the group with dynamic spasticity against 5 of 5 in the group without. A limb with calf overactivity is usually also a limb with a weak tibialis anterior, and a gait variable that tracks one will track the other through that association. Nor do the bedside instruments capture the phenomenon they are used to grade: using diagnostic tibial motor nerve block as the reference in 40 adults, [3] found no association between Tardieu-assessed spasticity and the soleus overactivity actually present during swing.

The one procedure that does supply a magnitude is the diagnostic block itself, which removes calf overactivity and leaves the weakness in place. It is invasive: it is the procedure the field would like to avoid, it removes only one of the two components, and it cannot calibrate the other. A simulation does not substitute for a patient study, and we do not offer it as one. It is, however, the only instrument in which both magnitudes are set rather than inferred, and in which a gait variable can therefore be asked what it carries about each. That is the question we put: not whether a stroke gait differs from an unimpaired one, which is settled, but which quantity a sagittal joint angle reports when both lesions are present and both are known. The variable this identifies is not a new measurement. Peak ankle dorsiflexion in swing is a standard output of three-dimensional gait analysis, reported with between-session reliability in post-stroke cohorts [15] and with population values in chronic stroke [5]. Nothing we propose requires a clinic to record anything it does not already record. The simulation supplies the calibration and not the variable: which component the number reports, how steeply, and (the part no patient study can establish) what it does not report at all.

S14. The dose response behind the sign reversal, and the ceiling on the rank correlation

This section carries the workings behind §3.4: the dose response that produces the sign reversal, the per-gain rank correlations, and the ceiling on a five-level correlation. Against the

530 unlesioned control, plantarflexor hyperreflexia displaces peak swing dorsiflexion by -9.161°
531 and dorsiflexor weakness displaces the same variable by -0.430° , a ratio of 21 to 1. That ratio
532 is particular to this variable, although the asymmetry itself is not. Across the 81 candidates the
533 median displacement is 0.49° for weakness against 1.90° for hyperreflexia, weakness reaches
534 3.95° at the knee in early swing, and on ten of the 81 it displaces more than hyperreflexia does.
535 The direction and the order of magnitude generalise. The factor of 21 does not, and the
536 reported endpoint is the candidate at which the asymmetry is most extreme.

537 The ankle at 75, 80 and 90 per cent of the cycle orders the six tibialis anterior scalings at $|\rho| \geq$
538 0.8 and over spans wider than one clinical measurement error, at $\rho = +0.822, +0.900$ and
539 -0.820 across 2.870, 2.208 and 2.272° : three of the 81. Applying the same test to the
540 hyperreflexia group, we find 25.

541 §3.4 reports two counts, of variables that usably grade calf overactivity and of variables that
542 usably grade dorsiflexor weakness. Neither is a fixed quantity, and two analytic choices move
543 them: which measurement error is used as the threshold, where four of the 81 variables have
544 published errors of their own and we had applied the smallest to all of them, and whether the
545 rank correlation uses mid-ranks on a tied dose vector, which the helper we first used did not.
546 Crossing the three defensible thresholds with the three rank conventions gives nine readings,
547 in which the tone count ranges from 14 to 28 and the weakness count from 1 to 7. Under the
548 reading this section's own sentence describes, an ordering of six scalings at the condition level
549 where no ties exist, they are 27 and 7. The ratio ranges from 3.7 to 14.0 across all nine and
550 never inverts. We therefore report the asymmetry as a direction and an order of magnitude,
551 roughly fourfold at its weakest and fourteenfold at its strongest, and not as the pair 25 and 3.
552 Supplement S8 gives the nine counts, the three thresholds and the three rank conventions
553 behind them.

554 The denominator is not 81 independent opportunities. We sampled three joint-angle curves
555 every 5 per cent of the cycle and added eighteen landmarks, so adjacent samples of one curve
556 are strongly correlated. Halving the sampling interval would roughly double both counts and
557 change nothing. Therefore, the counts index how much of the cycle each lesion is legible over
558 rather than how many separate measurements report it. On that reading calf overactivity is
559 legible over between a sixth and a third of the sagittal variables a gait laboratory produces, at
560 the ankle throughout swing and late stance and at the knee through loading response, and
561 weakness over a small fraction of that and far more faintly. The finding therefore does not
562 depend on our having chosen the right variable.

563 The three variables that grade weakness grade tone more strongly at the same phase. Their
564 hyperreflexia spans are 6.308, 6.055 and 4.148° against weakness spans of 2.870, 2.208 and
565 2.272°, 2.2, 2.7 and 1.8 times larger taking the three in the order listed. A reading taken there
566 moves for two reasons at once and the two cannot be separated from that reading alone.
567 Weakness depth is not recovered from a reading that does not already know the tone. On the
568 mixed grid it is recovered neither by knee angle at contact, nor by peak swing dorsiflexion,
569 nor by the two entered together, nor by the two with their interaction: every leave-one-
570 weakness-level-out coefficient of determination is negative. Each does worse than reporting
571 the mean. The fold is demanding but passable, since the parallel leave-one-gain-level-out
572 folds return R^2 between 0.64 and 0.82 for reflex gain, and the negative values are therefore
573 evidence about the weakness signal and not an artefact of having only four levels. Reflex gain,
574 by contrast, is recovered from one reading of one variable at Spearman $\rho = -1.000$ across the
575 five gain levels and -0.964 over the 23 hyperreflexia runs, with a leave-one-condition-out
576 calibration error of 0.0062 in delivered gain, 4.5 per cent of the series' range. That correlation
577 should be read with its ceiling in view: the series has five levels and the five condition means
578 are strictly ordered. Therefore, the condition-level ρ is 1.00 by construction, and what the run-

level figure adds is that the ordering survives seed-to-seed dispersion, and the calibration error that it survives holding a whole condition out. We ran that regression on four feature sets and not on all 81. One two-stage estimator was also tested on the same grid: grading delivered gain from peak swing dorsiflexion recovers it at $R^2 = 0.946$ under leave-one-run-out, and regressing the knee-at-contact residual on weakness afterwards recovers $R^2 = 0.129$. The first figure is not comparable to the 0.640 to 0.821 above, which are leave-one-gain-level-out and therefore a harder fold. What remains untested is the sign-appropriate variant, which would apply a different weakness relation on each side of the reversal. The statement that holds across all 81 is the displacement asymmetry above and the direction of the count, which survives every bar and rank convention tested even though its value does not.

The failure appears to have a mechanism behind it. The reported endpoint orders the weakness series as precisely as anything in the scan, $\rho = +1.000$ across the six scalings and $+0.966$ across cells, but over a span of 0.66° , and §3.5 sets that against the two measurement errors published for the variable. What the ordering loses in the presence of hyperreflexia is not its accuracy but its sign. Holding reflex gain fixed on the mixed grid and varying tibialis anterior scaling, the four weakness levels are ordered perfectly at delivered KV 0.025 ($\rho = +1.000$, span 1.201° , adjacent levels run-disjoint), flat at KV 0.050 ($\rho = -0.400$, span 0.191°), and ordered perfectly again with the opposite sign at KV 0.100 ($\rho = -1.000$, span 1.449° , again run-disjoint) and at KV 0.220 ($\rho = -1.000$, span 0.753°). The spans at the two ends exceed the 0.66° of the pure group. Whether weakening the dorsiflexors raises or lowers peak swing dorsiflexion depends on how hard the plantarflexors are pulling against them, and the crossover falls near delivered KV 0.050. An estimator that does not already know the tone cannot use a signal whose sign it cannot predict, which is why the regression fails and why the failure says what a working estimator would need.

S15. The transfer to a patient cohort, in full

604 §4.1 states the standardised effect and the misassignment rate; this section gives their
605 derivation, the cohort they are built from, and the caveats attaching to each denominator.
606 Against the spread a chronic stroke population presents at this landmark the picture changes.
607 Ref [5] splits 34 chronic stroke patients by initial-contact type and reports peak swing ankle
608 angle of $2.76 \pm 4.20^\circ$ in the forefoot-contact subgroup and $9.99 \pm 4.21^\circ$ in the rearfoot
609 subgroup (that table labels the row plantarflexion, while the paper's text reports the rearfoot
610 group as the more dorsiflexed. The values and the text agree with the reading used here).
611 Those subgroup SDs are conditioned on a split closely related to the quantity being predicted.
612 We take the whole-cohort SD as the denominator, which restores the 7.23° between-subgroup
613 mean difference the split removes. That figure, 5.55° , is computed here from the two
614 subgroup means and SDs, assuming the two subgroups are of equal size, and is not reported in
615 [5]. The same construction applied to knee angle at contact in the same cohort gives 8.57° ,
616 and the two are set against each other here. The cohort it comes from excluded 67 of 116
617 screened patients for walking with an orthosis or a walking device. Therefore, it is the aid-free
618 tail of the spastic population and probably its mildest. That cuts both ways: a wider cohort
619 would raise the denominator and lower d , and it is also the population in which a barefoot
620 recording of this variable is possible at all. §4.1 combines these with the measurement error to
621 give the total SD, the standardised effect and the misassignment rate. The reference set of 138
622 able-bodied adults [25] reports no swing-phase ankle SD and so cannot corroborate it. The
623 corresponding figure for knee angle at contact is 38.4 per cent, computed on that variable's
624 own between-patient SDs of 7.93 and 9.16° [5] and its own single-measurement error of 2.62°
625 [13]. That figure carries a caveat the ankle's does not: ref [13] measured 50 healthy subjects.
626 Therefore, the knee comparator's error is a healthy-subject reliability applied to a stroke
627 cohort, where ref [15] is post-stroke. Neither kind of number substitutes for the other, and
628 only the second bears on a single patient. A two-group comparison at the ankle would

629 therefore need roughly a sixth the sample it needs at the knee. That ratio is a property of the
630 discriminability and not a sample size for the study §4.3 proposes, which is correlational and
631 needs its own arithmetic; what transfers is that the choice of instant decides whether a patient
632 study is affordable.